E-Commerce Sales & Customer Journey Analysis

1. Project Overview and Objective

Project Overview

This project analyzes an e-commerce company's website traffic, customer behavior, product performance, sales, and refund data using Excel, Power Query, Power BI, and DAX. The objective is to transform raw transactional data into meaningful business insights through interactive dashboards.

Objective

- Analyze customer purchase behavior.
- Measure sales, revenue, profit, and refunds.
- Evaluate marketing campaign performance.
- Track website traffic and conversion rate.
- Build an interactive Power BI dashboard to support business decisions.

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2. Data Sources

Sources

- **Source Description and Timeline:** MavenAnalytics – Maven Fuzzy Factory Dataset
- Google Dataset Search **and** 2012-2015.
- **Domain**: E-Commerce / Retail Analytics, Dataset Tables, Website Sessions, Website Pageviews, Orders, Order Items, Products, Order Item Refunds.

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3. Problem Statement

The company wants to identify the factors affecting sales performance and customer conversions. The project aims to analyze website traffic, marketing campaigns, product sales, customer behavior, and refund trends to improve revenue and profitability.

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4. Attribute (Column /Features) Details:

No.	Data Type	Description
1	Website Sessions	Customer visit details, traffic source, campaign, device
2	Website Pageviews	Customer navigation across website pages
3	Orders	Order details, revenue, product purchased
4	Order Items	Product-level order information
5	Products	Product master details

6	Order Item Refunds	Refund transactions

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5. Tools & Technologies

- Excel – Data Cleaning & Pivot Tables
- Power Query – Data Transformation & Handling Missing Values
- Power BI – Data Modeling & Dashboard
- DAX – KPIs & Calculated Measures

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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- Removed duplicate records.
- Checked and handled missing values.
- Converted columns into proper data types.
- Created relationships between tables.
- Standardized date and text formats.
- Created Pivot Tables for preliminary analysis.
- Built Fact and Dimension tables.
- Imported cleaned data into Power BI.
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7. Data Modelling and DAX (Power BI)

Data Model:

Created relationships between the following tables:

- orders → order_items (Order ID)
- order_items → order_item_refunds (Order Item ID)
- website_sessions → website_pageviews (Website Session ID)

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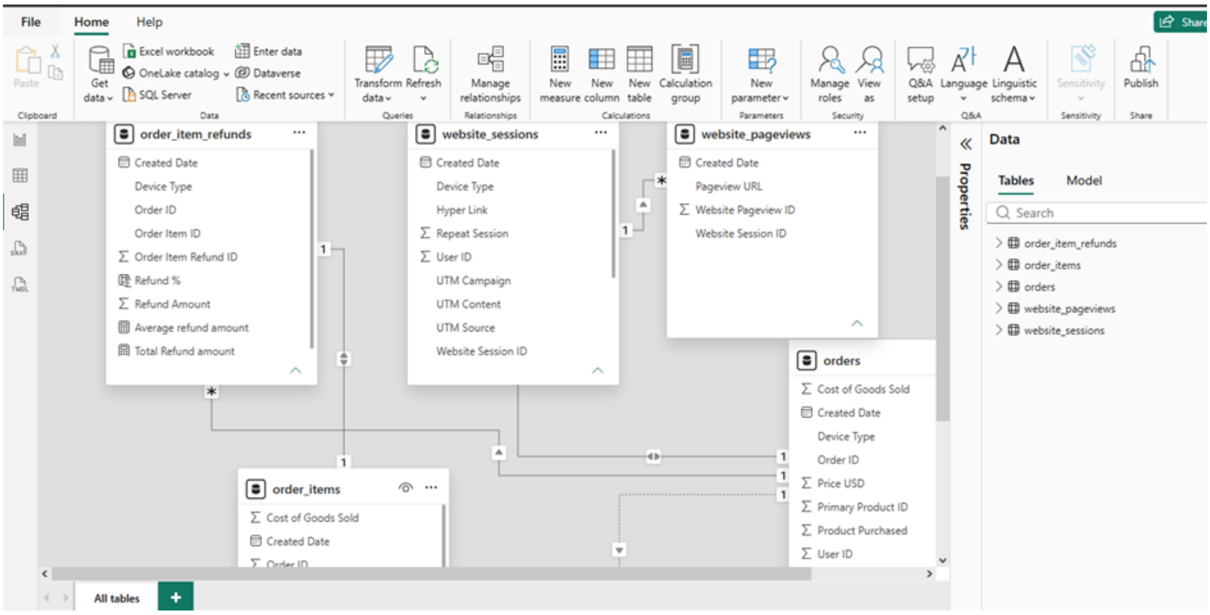


Calculated Columns & DAX Measures:

Relationship:

- One-to-Many (1:*)
- Cross Filter Direction: Single
- (Insert Screenshot of Model View here)
- Calculated Columns & DAX Measures (List Out)

Measures:

- `Average refund amount = AVERAGE(order_item_refunds[Refund Amount])`
- `Refund % = DIVIDE(SUM(order_item_refunds[Refund Amount]),SUM(orders[Price USD]),0)`
- `Total Refund amount = SUM(order_item_refunds[Refund Amount])`
- `Average Price = AVERAGE(order_items[Price USD])`
- `Average Profit = AVERAGE(order_items[Profit])`
- `Net Revenue = [Total Revenue]-order_item_refunds[Total Refund amount]`
- `Profit = DIVIDE(SUM(order_items[Cost of Goods Sold]),SUM(order_items[Price USD]),0)`
- `Total COGS = SUM(order_items[Cost of Goods Sold])`
- `Total Price USD = SUM(order_items[Price USD])`
- `Total product Count = COUNT(order_items[Product ID])`
- `Total Revenue = SUM(order_items[Profit])`
- `New custotmers = CALCULATE(SUM(website_pageviews[Website Session ID]),website_sessions[Repeat Session])=0`
- `Repeated custotmers = CALCULATE(SUM(website_pageviews[Website Session ID]),website_sessions[Repeat Session])`
- `Sessions = COUNT(website_sessions[User ID])`
- `Total Sessions = COUNTROWS(website_sessions)`
- `User = DISTINCTCOUNT(website_sessions[User ID])`

8. Analysis and Visualizations (Power BI)

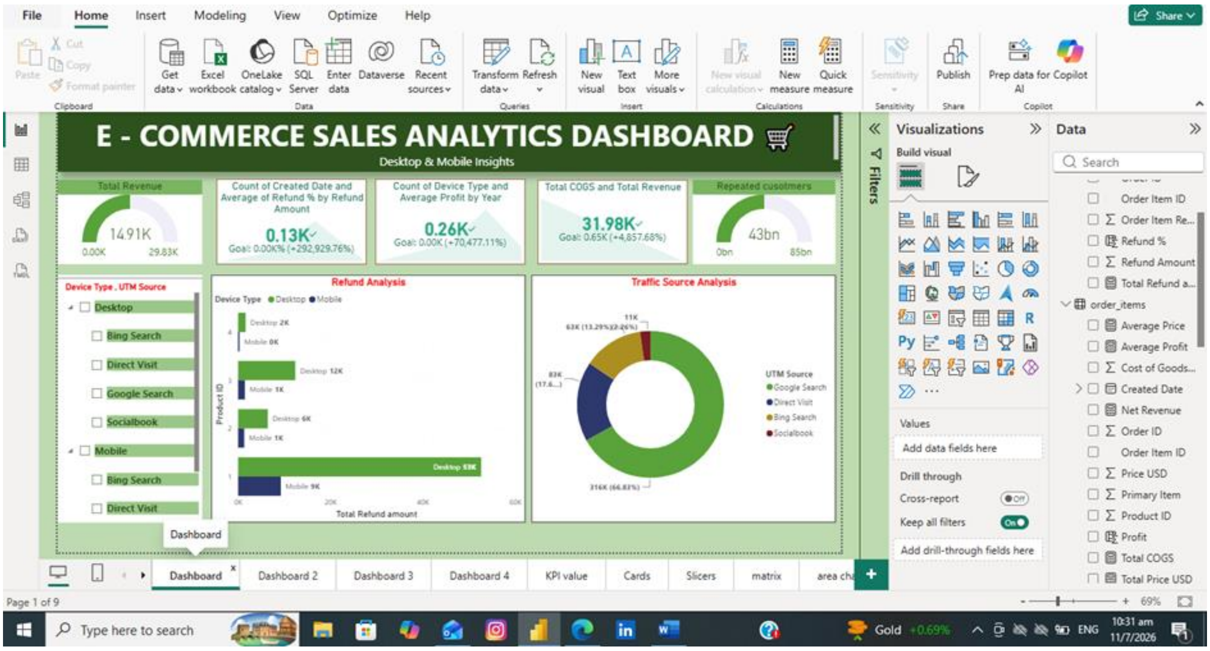
Dashboard Features:

Visualizations used:

- KPI Cards
- Total Revenue
- Total COGS
- Total Refund Amount
- Profit Margin %
- Average Price
- Total Sessions
- Bar Chart
- Refund Amount by Product
- Line Chart
- Monthly Revenue Trend
- Donut Chart
- Pie Chart
- Sessions by Device Type
- Matrix
- Revenue by Year and Device
- Treemap
- UTM Campaign Analysis
- Slicers
- Year
- Quarter
- Device Type
- UTM Source

Interactive Features:

- ✔ Drill Down
- ✔ Cross Filtering
- ✔ Slicers
- ✔ Bookmarks
- ✔ Clear Titles
- ✔ Data Labels
- ✔ Consolidated Dashboard


E - COMMERCE SALES ANALYTICS DASHBOARD
Desktop & Mobile Insights
Total Revenue
14.91K
Count of Created Date and Average of Refund % by Refund Amount
0.13K
Count of Device Type and Average Profit by Year
0.26K
Total COGS and Total Revenue
31.98K
Repeated customers
43bn
Device Type , UTM Source
Desktop
Bing Search
Direct Visit
Google Search
Socialbook
Mobile
Refund Analysis
Total Refund amount
Traffic Source Analysis
UTM Source
Google Search
Direct Visit
Bing Search
Socialbook
Visualizations
Data
Dashboard
Dashboard 2
Dashboard 3
Dashboard 4
KPI value
Cards
Slicers
matrix
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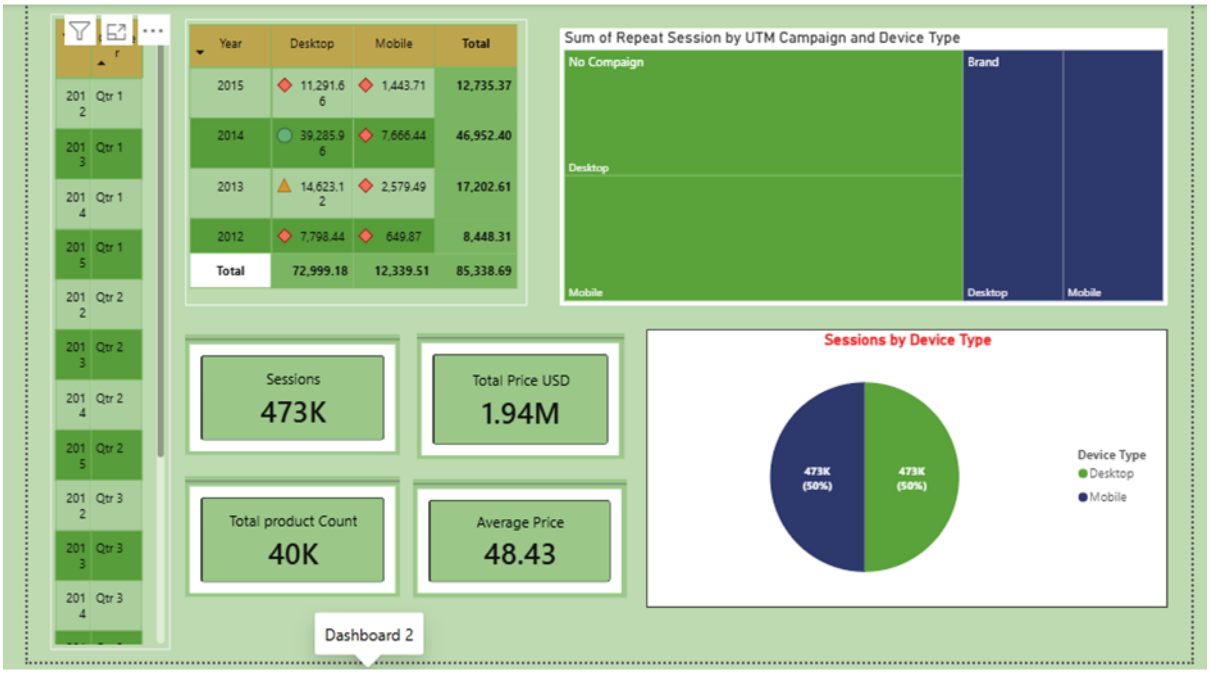
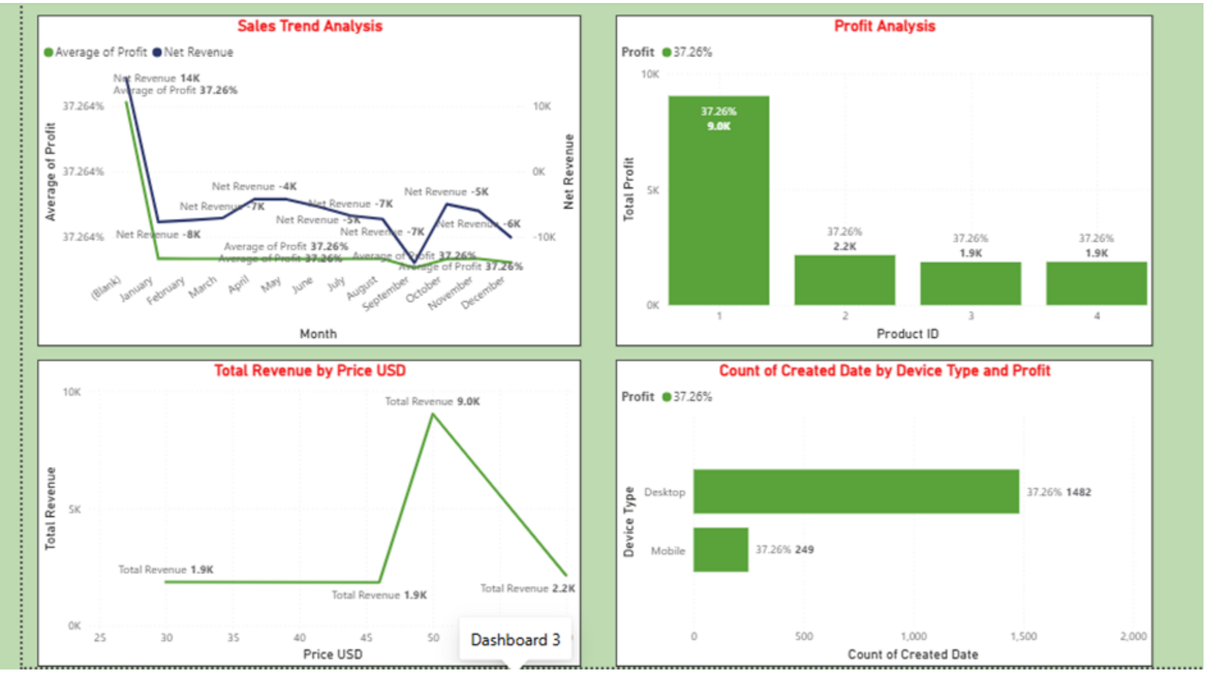

Year
Desktop
Mobile
Total
2015
11,291.6 6
1,443.71
12,735.37
2014
39,285.9 6
7,666.44
46,952.40
2013
14,623.1 2
2,579.49
17,202.61
2012
7,798.44
649.87
8,448.31
Total
72,999.18
12,339.51
85,338.69
Sum of Repeat Session by UTM Campaign and Device Type
No Compaign
Brand
Desktop
Mobile
Sessions
473K
Total Price USD
1.94M
Total product Count
40K
Average Price
48.43
Sessions by Device Type
473K (50%)
Device Type
Desktop
Mobile
Dashboard 2



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9. Insights & Conclusions

Key Findings:

Desktop users generated higher revenue than Mobile users.
Google Search contributed the highest traffic.
Refund percentage was relatively low compared to total revenue.
Profit margin remained around 37.26%.
Revenue peaked during specific months, showing seasonal trends.
Average product price was approximately 48 USD.

Analysis insights:

- Summarized sales, revenue, profit, refunds, and customer sessions using KPI cards and charts.
- Identified that Desktop traffic and Google Search generated the highest revenue, while refund amounts were concentrated on specific products
- Based on historical sales trends, future revenue is expected to increase during peak months if traffic continues to grow.
- Improve Mobile conversion rate
- Focus more on Google Search campaigns
- Reduce refunds by improving product quality
- Increase marketing for high-profit products

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10. Conclusions

This project demonstrates how Excel and Power BI can transform raw e-commerce data into meaningful business insights. Through data cleaning, modelling, DAX calculations, and interactive dashboards, key business metrics such as Revenue, Profit, Refunds, Customer Sessions, and Traffic Sources were analyzed. The dashboard enables stakeholders to make informed decisions and improve business performance.